core of such vehicles is the numerous silicon products, as they are crucial enablers for all the alternate fuel (non-internal combustion engines) technologies where efficiency management becomes critical.

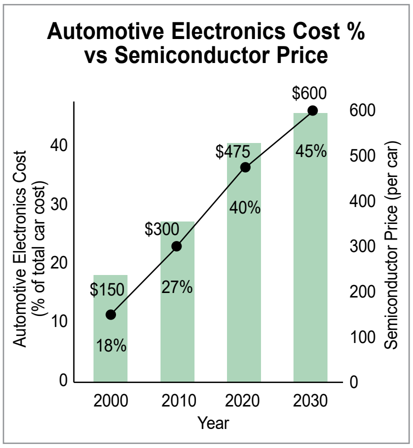
Furthermore, the Indian government's push towards establishing a comprehensive electric charging infrastructure and its plan to ensure that 30% of all vehicles on the road are electric by 2030 also shows how crucial semiconductor components will be, which are, by default, needed to enable intelligent charging infrastructure.

Indian automotive semiconductor landscape

Modern vehicles require anywhere from 500 to 1500 semiconductor chips per vehicle. Advancements in safety features, infotainment systems, and autonomous driving technologies, among other things, drive this demand.

As vehicles become more sophisticated with features like advanced driver assistance systems (ADAS), electric vehicles (EVs), and connected car technologies, the demand for semiconductors in the automotive sector is surging. This trend will continue to gain traction, with semiconductor content per car projected to increase significantly, highlighting the sector's importance in driving the future of automotive technology.

Despite the ambitious plans and the substantial growth of India's automotive industry, the country faces significant challenges in automotive semiconductor manufacturing, a critical component of modern automotive technology. As of now, India's efforts in semiconductor manufacturing are in the nascent stages, with the country working towards establishing its first semiconductor fabrication plants (fabs) capable of producing chips with 28nm technology. This technology, while a leap forward for



(Data Source: Deloitte)

India's semiconductor ambitions, is still behind the cutting-edge processes used in leading semiconductor manufacturing countries, where fabs can produce chips as small as 5nm or even 3nm, which even automotive semiconductor companies have started to adopt. India's automotive industry can still benefit from Tata-PSMC's 28nm technology, which will require several years of close coordination with India-based automotive companies.

Until then, the Indian automotive industry entirely relied on importing semiconductor automotive products, making it, among other sectors, vulnerable to global supply chain disruptions, as vividly demonstrated during the Covid-19 pandemic. The global chip shortage highlighted the precarious position of industries reliant on semiconductors, with many automotive manufacturers forced to cut production, leading to significant financial losses and delays in delivering new vehicles.

Therefore, the establishment of a domestic semiconductor manufacturing facility catering to the automotive industry is not just a matter of industrial policy but a strategic necessity for India's two major sectors: automotive and semiconductor. This is especially true as semiconductors are becoming more crucial for supporting the automotive sector's growth ambitions and ensuring its resilience against global supply chain vulnerabilities.

Cost of automotive semiconductor

The rising cost of semiconductor content in automotive electronics is another angle that shows how crucial it is to the growth trajectory of India's automotive industry.

Like the global shift towards electric, autonomous, and technologically advanced vehicles, the Indian market is also experiencing increased demand for vehicles equipped with advanced safety, infotainment, connected solutions, 360-degree view cameras, and many other driving assistance systems. This evolution elevates the role of semiconductors, the critical components at the heart of these electronic systems, thereby increasing the cost per vehicle.

From 2013 to the present, the semiconductor content cost per vehicle has significantly risen, jumping from approximately US$312 to around US$400, and projected to approach US$600 by 2024. This increase is indicative of the greater quantity of semiconductors required per vehicle and the industry's move towards more sophisticated technologies-driven trends. These systems require high-end processors, sensors, and power control units, emphasising the shift towards higher performance and more complex semiconductor devices.

For the Indian automotive industry, these rising costs present both challenges and opportunities. On the challenge side, the increased cost of semiconductors contributes to higher overall vehicle prices, which could impact consumer affordability and the rate of technology adoption. However, this also presents a significant opportunity for Indian manufacturers to distinguish their offerings through cutting-edge technology, potentially